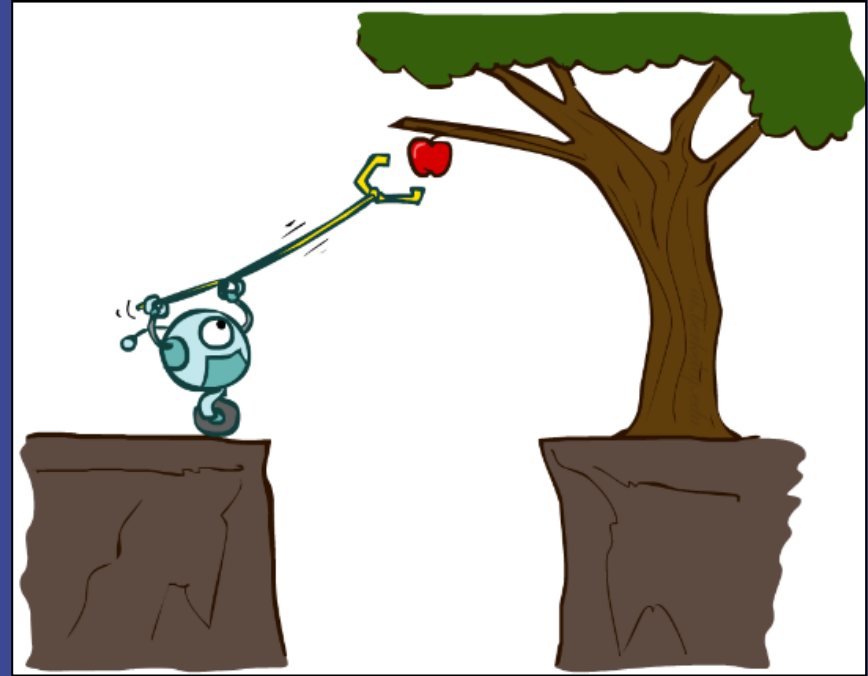


CS 156

Intro to AI

Problem Solving Agents



Some slides and demos are based on the slides created by Dan Klein and Pieter Abbeel for CS188 Intro to AI at UC Berkeley.

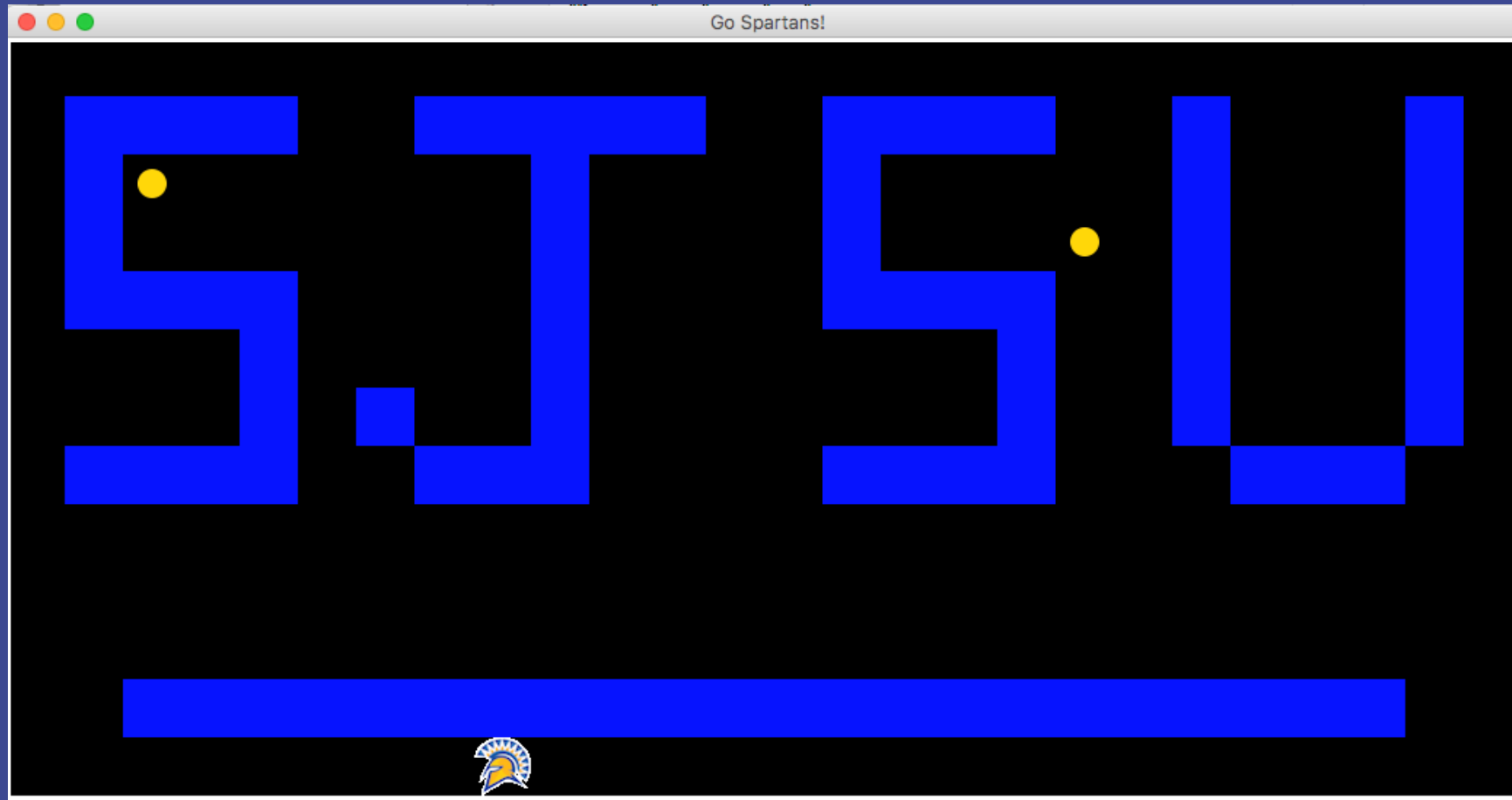
The artwork is by Ketrina Yim

Problem Solving Agents

A problem solving agent is a special type of **goal based agent** operating in an environment that is:

- ▶ fully observable
 - ▶ discrete
 - ▶ deterministic
 - ▶ single agent
-
- ▶ A problem solving agent uses an **atomic representation** of the world.
 - ▶ A problem solving agent uses **search** to find a solution.
 - ▶ The solution is a **sequence of actions** that will achieve the goal.

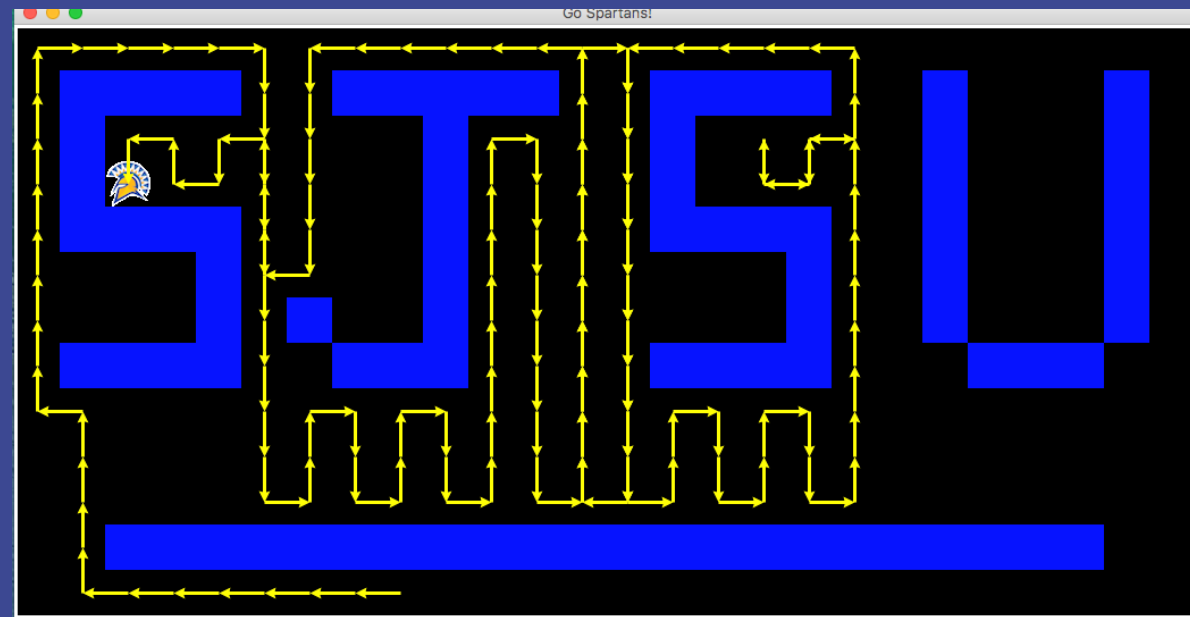
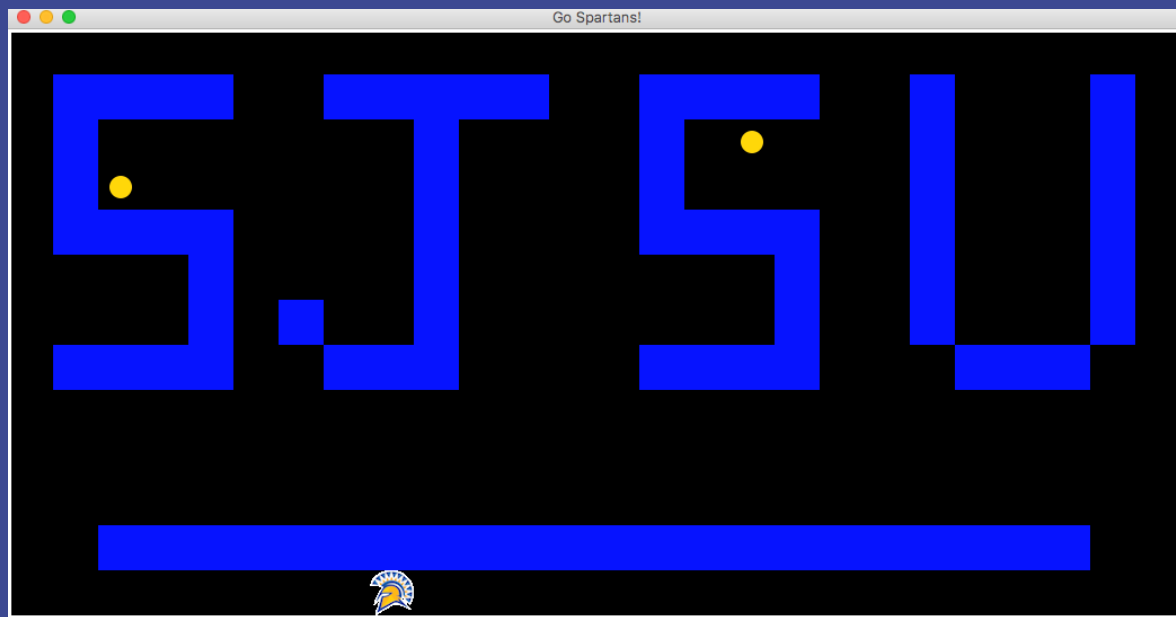
Problem Solving Agent



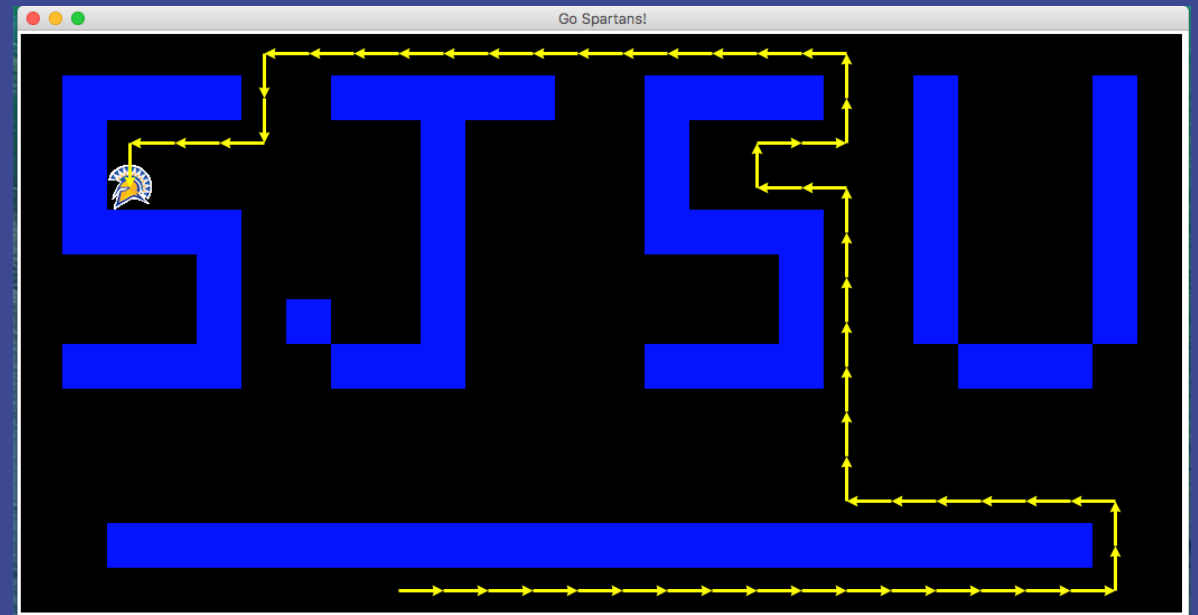
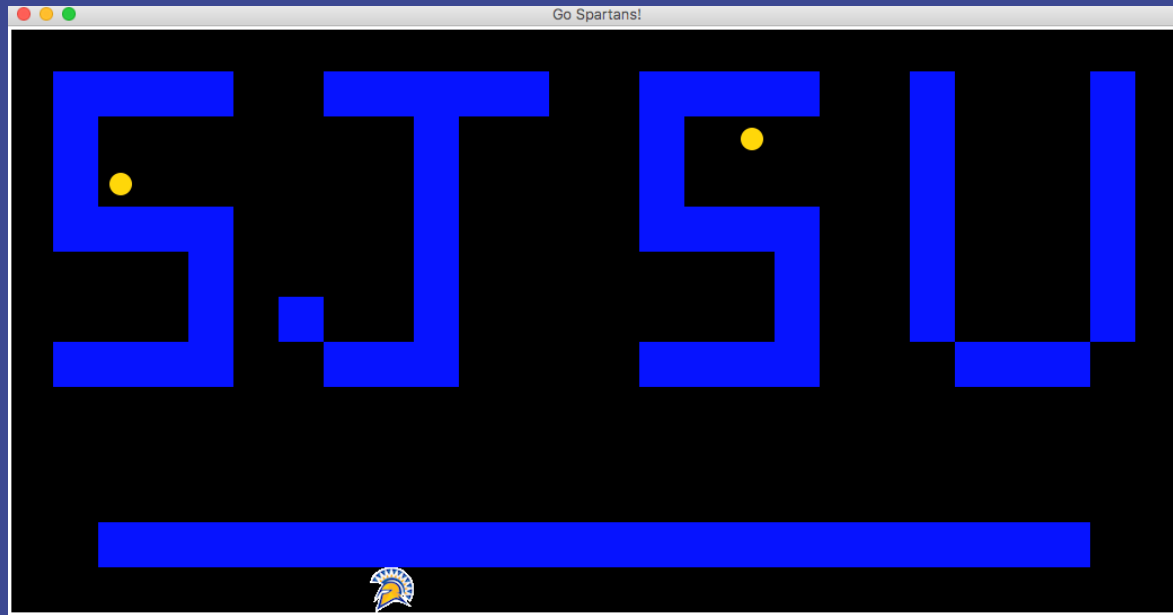
Evaluating Problem Solving Agents

- ▶ Optimal vs. complete
 - Complete: the agent is guaranteed to find a solution
 - Optimal: the agent finds the best solution

Optimal vs Complete

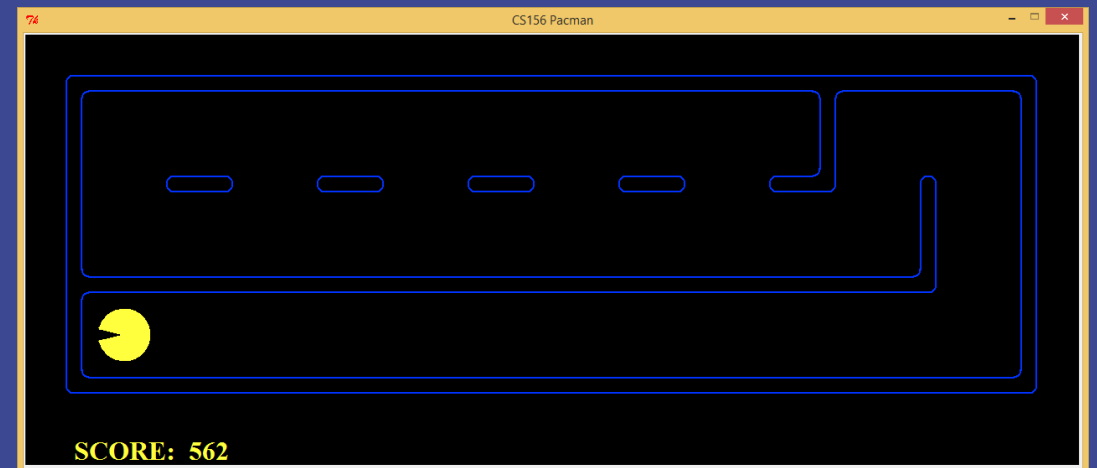
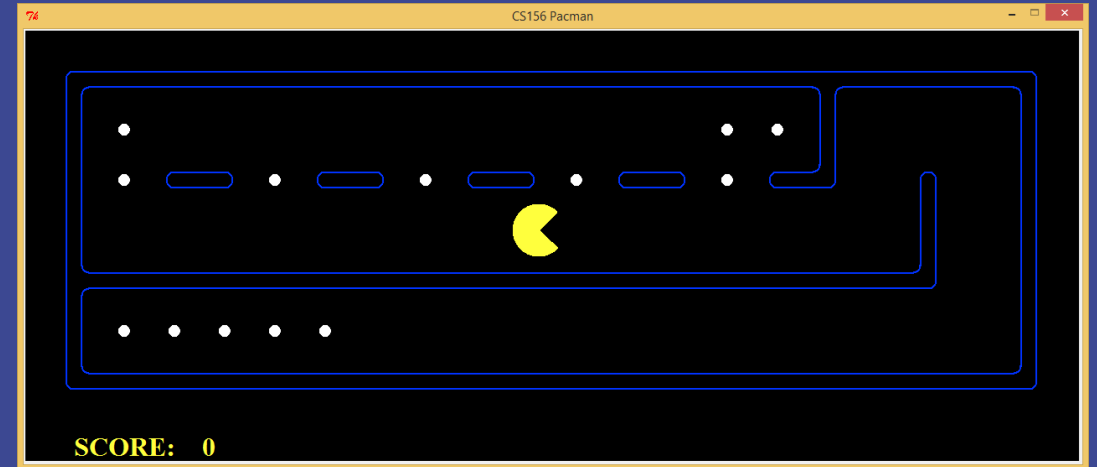


Optimal vs Complete



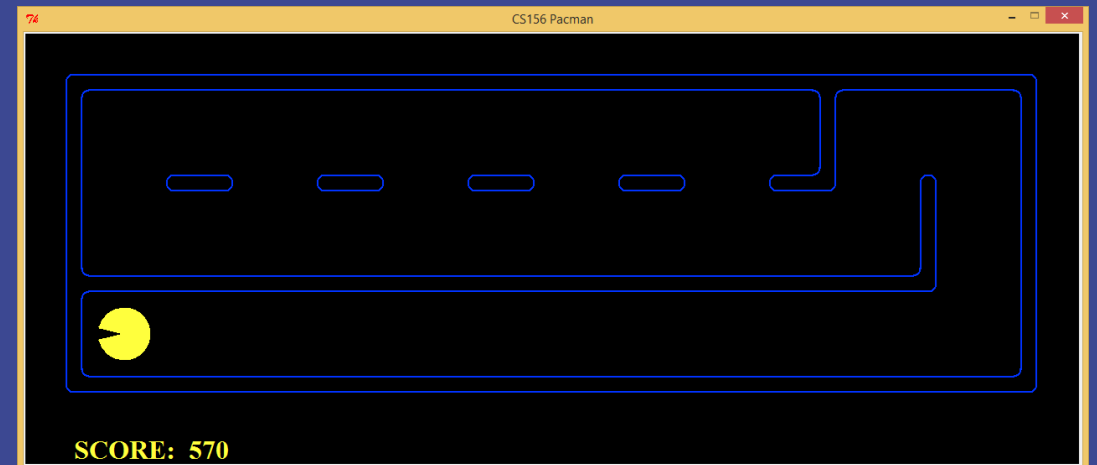
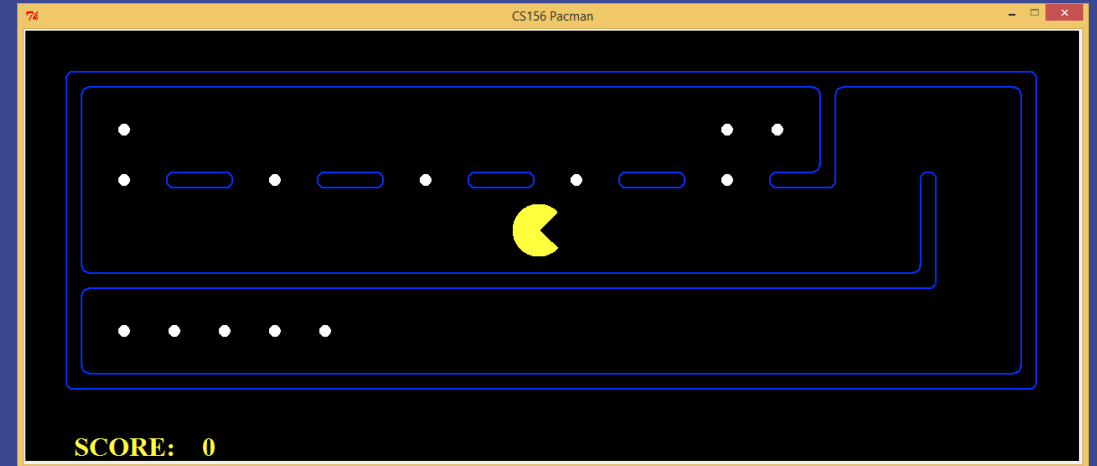
Pacman Demo 1

- ▶ It comes up with mini-plans and executes them one after the other
- ▶ Very fast
- ▶ It is **complete**: it found a way to achieve its goal of clearing the dots
- ▶ May not be the most efficient way of clearing the dots



Pacman Demo 2

- ▶ Spends ~ 2 seconds searching for the best way to clear all the dots
- ▶ Not a single wasted step
- ▶ It is **complete**: it found a way to achieve its goal of clearing the dots.
- ▶ It is **optimal**: found the 'best' way to achieve its goal



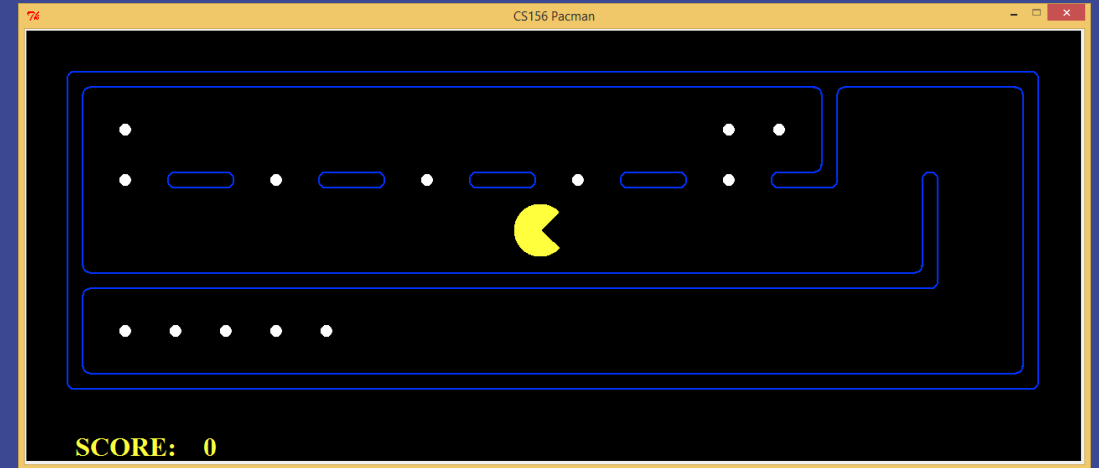
Problem Solving Agent Design

1. Formulate the problem
2. Search for a solution
3. Execute the solution

Search Problems

Define the goal:

- ? Eat the closest dot
- ? Eat a dot
- ? Eat all the dots
- ? Eat as many dots as possible within a given time



Search Problems

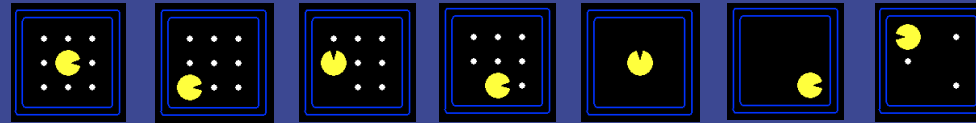
A search problem consists of:

- A state space: a set of possible states that the environment can be in. Each state describes **how the world is at a given time.**

Search Problems

A search problem consists of:

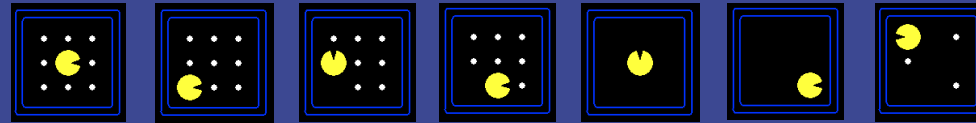
- A state space:



Search Problems

A search problem consists of:

- A state space:

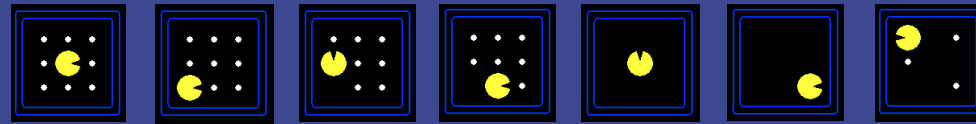


- A successor function: an abstraction that **models** how we think the world works, how it evolves in response to given actions. It includes actions and costs. It represents our **transition model**.

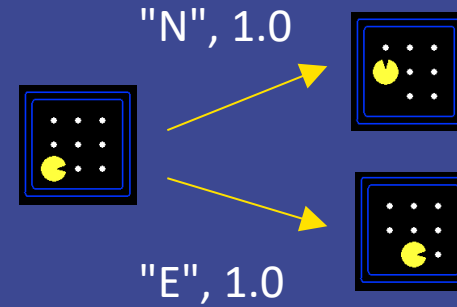
Search Problems

A search problem consists of:

- A state space:



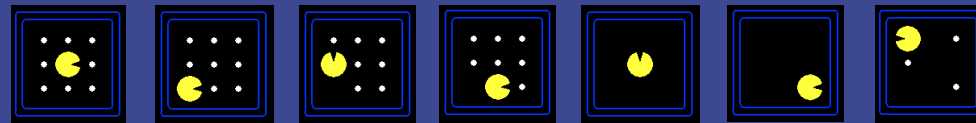
- A successor function:
(with actions, costs)



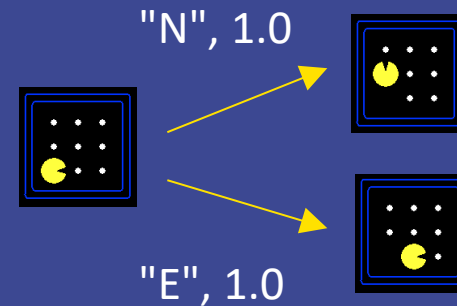
Search Problems

A search problem consists of:

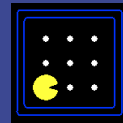
- A state space:



- A successor function:
(with actions, costs)



- A start state:



- A goal test: all the dots are eaten? Pacman position?

Search Problems

- ▶ A solution is a **sequence of actions** (a plan) which takes us from the start state to a state that satisfies the goal test.

